

Neymanian Repeated Sampling Inference in Completely Randomized Experiments

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Finite population setup

- FRT focuses on individual treatment effect
- Neymanian inference considers average causal effect (Neyman, 1923, 1990)

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- FRT focuses on individual treatment effect
- Neymanian inference considers average causal effect (Neyman, 1923, 1990)
- Potential outcomes: $Y_i(1)$ and $Y_i(0)$ for $i = 1, \dots, n$
- Individual effect: $\tau_i = Y_i(1) - Y_i(0)$
- Treatment $\mathbf{Z} = (Z_1, \dots, Z_n)$
- Completely randomized experiments

$$\text{pr}(\mathbf{Z} = \mathbf{z} \mid \mathbf{Y}(1), \mathbf{Y}(0)) = \begin{cases} \frac{1}{\binom{n}{n_1}} & \text{if } \sum \mathbf{z}_i = n_1 \\ 0 & \text{if } \sum \mathbf{z}_i \neq n_1 \end{cases}$$

- Finite population inference: randomness comes solely from $\mathbf{Z}$

Finite population quantities

- Finite population means

$$\bar{Y}(1) = \frac{1}{n} \sum_{i=1}^n Y_i(1), \quad \bar{Y}(0) = \frac{1}{n} \sum_{i=1}^n Y_i(0)$$

- Quantity of interest: $\tau = \frac{1}{n} \sum_{i=1}^n \tau_i = \bar{Y}(1) - \bar{Y}(0)$

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- Finite population variances

$$S^2(1) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\}^2, \quad S^2(0) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(0) - \bar{Y}(0)\}^2$$

$$S(1,0) = \frac{1}{n-1} \sum_{i=1}^n \{Y_i(1) - \bar{Y}(1)\} \{Y_i(0) - \bar{Y}(0)\}$$

- Variance of the individual effect

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- $2S(1,0) = S^2(1) + S^2(0) - S^2(\tau)$

Estimators

- Sample means

$$\hat{Y}(1) = \frac{1}{n_1} \sum_{i=1}^n Z_i Y_i, \quad \hat{Y}(0) = \frac{1}{n_0} \sum_{i=1}^n (1 - Z_i) Y_i$$

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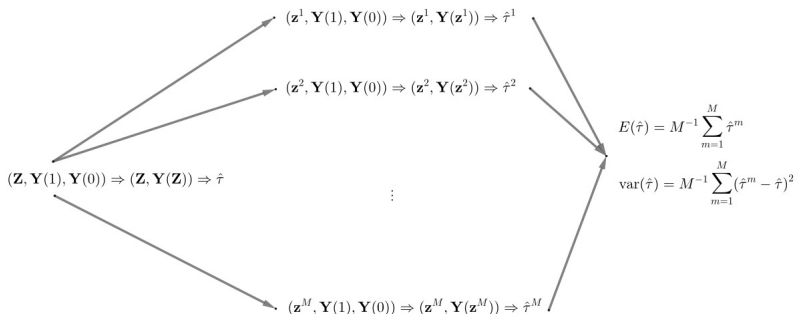
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- Difference-in-means estimator: $\hat{\tau} = \hat{Y}(1) - \hat{Y}(0)$

- Unbiasedness: $\mathbb{E}(\hat{\tau}) = \tau$

$$\mathbb{E}\left(\frac{1}{n_1} \sum_{i=1}^n Z_i Y_i\right) = \mathbb{E}\left(\frac{1}{n_1} \sum_{i=1}^n Z_i Y_i(1)\right) = \left(\frac{1}{n_1} \sum_{i=1}^n \mathbb{E}(Z_i) Y_i(1)\right) = \frac{1}{n} \sum_{i=1}^n Y_i(1)$$

Repeated sampling inference



- The expectation and variance are computed with respect to the finite population framework
 - potential outcomes are fixed
 - repeated sampling of Z_i under the CRE

Variance

- $\text{var}(\hat{\tau}) = \text{var}\{\hat{Y}(1)\} + \text{var}\{\hat{Y}(0)\} - 2\text{cov}\{\hat{Y}(1), \hat{Y}(0)\}$

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- A conservative 95% confidence interval based on normal approximation (Finite population CLT)

$$\left(\hat{\tau} - 1.96\sqrt{\widehat{\text{var}}(\hat{\tau})}, \hat{\tau} + 1.96\sqrt{\widehat{\text{var}}(\hat{\tau})} \right)$$

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 - $\hat{\beta} = \hat{\tau}$
 - but the OLS variance estimator from *lm* function in *R* is

$$\begin{aligned}\hat{V}_{\text{ols}} &= \frac{n(n_1 - 1)}{(n - 2)n_1n_0} \hat{S}^2(1) + \frac{n(n_0 - 1)}{(n - 2)n_1n_0} \hat{S}^2(0) \\ &= \frac{\hat{S}^2(1)}{n_0} + \frac{\hat{S}^2(0)}{n_1}\end{aligned}$$

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- We need to use the Eicker-Huber-White (EHW) robust variance estimator

Heteroskedasticity Robust Variance Estimator

- Eicker-Huber-White robust variance estimator:

$$\begin{aligned}\underbrace{\hat{V}_{\text{EHW}}(\hat{\alpha}, \hat{\beta})}_{\text{sandwich}} &= \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{\text{bread}} \underbrace{(\mathbf{X}^\top \text{diag}(\hat{\epsilon}_i^2) \mathbf{X})}_{\text{meat}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{\text{bread}} \\ &= \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \left(\sum_{i=1}^n \hat{\epsilon}_i^2 \mathbf{x}_i \mathbf{x}_i^\top \right) \left(\sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1}\end{aligned}$$

where $\mathbf{X}_i = (1, Z_i)^\top$ and $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)^\top$

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- HC2 variant of the EHW variance estimator is identical to $\hat{V}$
- R code `lm(Y ~ Z)` and `hccm(model, type='hc2')`

Reducing Transphobia (Brookman and Kalla, 2016, *Science*)

- Can perspective-taking – “imagining the world from another’s perspective” – reduce transphobia?
- Treatment group ($n_1 = 912$): canvasser asking people to think about a time when they were judged unfairly and were guided to translate that experience to a transgender individual’s experience
- Control group ($n_0 = 913$): canvasser asking people to recycle
- Outcome variable: feeling thermometer towards transgender people (0 – 100)
 - ① baseline
 - ② 3 day
 - ③ 3 week
 - ④ 6 week
 - ⑤ 3 month

Durable Effects

- 6 week effect
- Dropping additional observations with missing covariates $n_1 = 278$ and $n_0 = 287$
- Difference in means: est. = 5.99, s.e. = 2.29, 95% CI = [1.50, 10.48]
- Simple regression: est. = 5.99
 - s.e. under homoscedasticity: 2.29
 - s.e. without homoscedasticity: 2.29

Finite population inference vs. super population inference

- Super population inference: $Y_i(1)$ and $Y_i(0)$ are i.i.d. sampled from a super population
 - difference in means estimator
 - variance: $\frac{\text{var}\{Y_i(1)\}}{n_1} + \frac{\text{var}\{Y_i(0)\}}{n_0}$
- Finite population inference: $Y_i(1)$ and $Y_i(0)$ are fixed
 - difference in means estimator
 - variance: $\frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}$

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 - variance: $\frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n}$
- Two steps for generating the observed data in superpopulation: (a) sample $Y_i(1)$ and $Y_i(0)$ for $i = 1, \dots, n$ from a super population; (b) complete randomization within the sampled potential outcomes
 - finite population inference considers (b); super population inference considers (a)+(b)
 - $\text{var}_f(\hat{\tau}) = \text{var}(\hat{\tau} \mid \mathbf{Y}(1), \mathbf{Y}(0))$ and $\text{var}_s(\hat{\tau}) = \text{var}(\hat{\tau})$

Summary: Fisher vs. Neyman

- Like Fisher, Neyman proposed randomization-based inference
- Unlike Fisher,
 - ① estimands are average causal effects
 - ② heterogeneous treatment effects are allowed
 - ③ asymptotic approximation is required for inference
 - ④ conservative variance estimator
 - ⑤ CLT may fail with heavy-tailed outcome
- Reading: IMBENS AND RUBIN, CHAPTER 6